

Logic Building Assignment : 12

Calculate Time Complexity of each program.

1) Print all factors of a number

```
#include <stdio.h>

void print_factors(int number)
{
    // TODO: print all positive factors of 'number' in ascending order
}

int main()
{
    int number;

    printf("Enter number : ");
    scanf("%d", &number);
    print_factors(number);

    return 0;
}
```

2) Count total factors of a number

```
#include <stdio.h>

int count_factors(int number)
{
    // TODO: return the total number of positive factors of 'number'
    return 0;
}

int main()
{
    int number;

    printf("Enter number : ");

    scanf("%d", &number);
    printf("%d\n", count_factors(number));
}
```

```
    return 0;
}
```

3) Sum of all factors of a number

```
#include <stdio.h>
```

```
int sum_of_factors(int number)
{
    // TODO: return sum of all positive factors of 'number'
    return 0;
}
```

```
int main(void)
{
    int number;

    printf("Enter number : ");

    scanf("%d", &number);

    printf("%d\n", sum_of_factors(number));
    return 0;
}
```

4) Sum of even factors (exclude the number itself)

```
#include <stdio.h>
```

```
int sum_of_even_factors(int number)
{
    // TODO: return sum of all positive even factors excluding 'number'
    return 0;
}
```

```
int main()
{
    int number;

    printf("Enter number : ");

    scanf("%d", &number);
    printf("%d\n", sum_of_even_factors(number));
    return 0;
}
```

5) Check if number is divisible by 5

```
#include <stdio.h>
#include <stdbool.h>

bool is_divisible_by_five(int number)
{
    // TODO
    return false;
}

int main(void)
{
    int number;

    printf("Enter number : ");

    scanf("%d", &number);
    printf("%s\n", is_divisible_by_five(number) ? "Yes" : "No");

    return 0;
}
```

